PART 1 (Continued): ARQ PROTOCOL & SUMMARY COMPARISON

6. Automatic Repeat Requests (ARQ) — TB Pages 61–62

ARQ is a network protocol that uses positive acknowledgements (ACK), negative acknowledgements (NAK), and timeout limits to achieve reliable data transmission over unreliable channels.

- **Positive Acknowledgement (ACK):** The receiver sends an ACK message upon receiving a clean packet with valid checksum/parity.
- **Negative Acknowledgement (NAK):** The receiver sends a NAK message when packet corruption is detected, prompting immediate resend.
- **Timeout Timer:** The sender starts a countdown timer immediately upon sending a packet. If no ACK or NAK is received before the timer expires (due to a lost packet), the sender automatically re-transmits the identical data block.

Detection Method	How It Works	Primary Advantage	Major Limitation
Parity Check	Appends 1 parity bit to ensure even or odd number of 1s in byte.	Extremely fast; tiny overhead (1 bit per byte).	Cannot detect even-numbered bit flips in same byte.
Parity Block	2D matrix of row parity bits and a vertical parity byte.	Pinpoints exact bit coordinates to auto-correct errors.	Ineffective if multiple bits in the same row and column fail.
Checksum	Calculates mathematical sum of block bytes and appends value.	High reliability across large files and data blocks.	Requires computational processing at both ends.
Echo Check	Receiver transmits copy of received data back to sender.	Simple implementation without complex math.	Doubles network traffic; cannot localize error direction.
Check Digit	Appended final digit calculated from prior digits using weights.	Detects human input, transposition and transcription errors.	Only used for manual entry/barcodes, not network packets.
ARQ Protocol	Combines error detection with ACK, NAK, and timeout resends.	Guarantees 100% complete data delivery across noisy channels.	Introduces latency while waiting for timeouts or re-sends.